Indianapolis Service Request Response Analysis

Where responses are slowest, when demand peaks, and what the city should plan for

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DATE
October 2026

Portfolio project: simulated client engagement using public City of Indianapolis data.

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OVERVIEW

Executive summary

This report analyzes Indianapolis 311 service requests handled through the Mayor's Action Center. It looks at which request types and areas wait longest for a response, when different request types peak during the year, and what call volumes the city should expect. The goal is to help the city plan staffing and projects so that service requests are resolved faster.

4.0 yrs
Longest median wait, for broken sidewalks and curbs

50%
Share of all chuckhole cases that arrive in January through March

19.7%
Overall error of the Q1 call forecast when tested against actual results

Key findings

1. **Request type matters more than location.** Within each request type, response times look similar across ZIP codes. Areas that wait longer appear to do so because they generate more of the slow request types, such as trash and street issues, rather than because of where they are.
2. **Sidewalk and street surface requests wait longest.** The four slowest request types have median waits between 2.9 and 4.0 years.
3. **Demand is strongly seasonal.** Half of chuckhole cases arrive from January to March. Weeds on private property peak from April to September, with 56% of cases in April to June. Trash-related requests are spread much more evenly across the year.
4. **A simple seasonal average makes a workable planning baseline.** Council districts 11 and 17 lead the forecast for all three of the most common request types, and district 2 leads for chuckholes. Treat the projected counts as a floor, since actual calls ran higher than projected.

Recommendations at a glance

- Schedule street maintenance in December to prepare for the January to March chuckhole peak, focusing on council districts 2, 11, 1, 9, and 17.
- Add capacity and homeowner support for weeds and private property complaints ahead of April to June.
- Plan trash response (dumping and stolen carts) around council districts 17, 12, and 11.
- Target the longest waits: broken and new sidewalk and uneven street surface requests.

CONTEXT

Background and approach

Data

The analysis uses the City of Indianapolis Mayor's Action Center service cases from the legacy case system (June 2015 to August 2022), published on the OpenIndy data portal. The dataset holds 1,220,121 cases, of which the 936,393 created before the August 2022 cutoff are analysed. Each case records the request type, location, and the dates it was created and closed. Response time is the time between creation and closure, so the response time analysis uses closed cases, while the seasonal and forecast analyses include both open and closed cases.

What was included and excluded

- **The analysis covers cases created before August 1, 2022.** The city changed its intake system around that date and later records are largely missing a subcategory, so they cannot be assigned a request type. The cutoff removes 283,728 of 1,220,121 cases (23.3%).
- **Request-for-information and switchboard cases were excluded.** These are quick informational calls that close almost immediately and do not require a crew to respond. Including them would pull the typical response time down and hide the delays that matter.
- **Rows with missing values were excluded from the analyses that need them.** After the cutoff this affects very little of the data; see below.
- **Minimum case counts were applied.** Request types needed at least 300 closed cases to be ranked and ZIP codes at least 84 to appear in the heatmap, so small groups do not distort the results.

Measures

- **Median response time** is used instead of the average, because a few very long cases would otherwise inflate the result.
- **Geography** is ZIP code for the response time analysis and council district for the forecast.

Missing values

Within the analysis window, subcategory and ZIP code are complete for every remaining case, so no request type or ZIP level result loses data to missing values. Council district is missing for about 2% of closed cases, most of it a gap between April and May 2019; those cases are excluded from the district forecast in Question 3. Ten cases are missing a created or closed date and are excluded everywhere.

QUESTION 1

Where and for what are city service responses slowest?

> ***Client question.*** *Which areas and request types experience the longest response times, and which combinations of the two have the longest delays?*

The slowest request types

Among request types with at least 300 closed cases, sidewalk and street surface repairs, new sidewalks, graffiti, and brush removal have the longest median waits. Broken sidewalks and curbs wait the longest, at a median of 4.0 years.

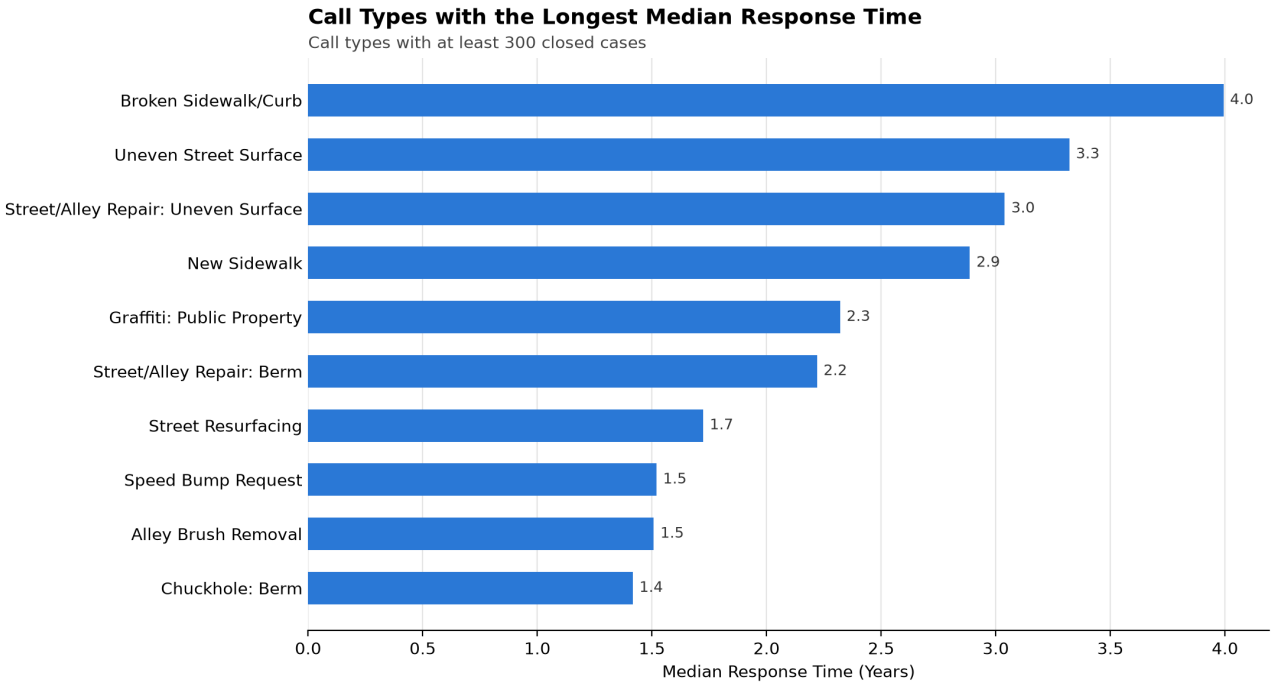


Figure 1. Request types with the longest median response time (minimum 300 closed cases).

Request type versus location

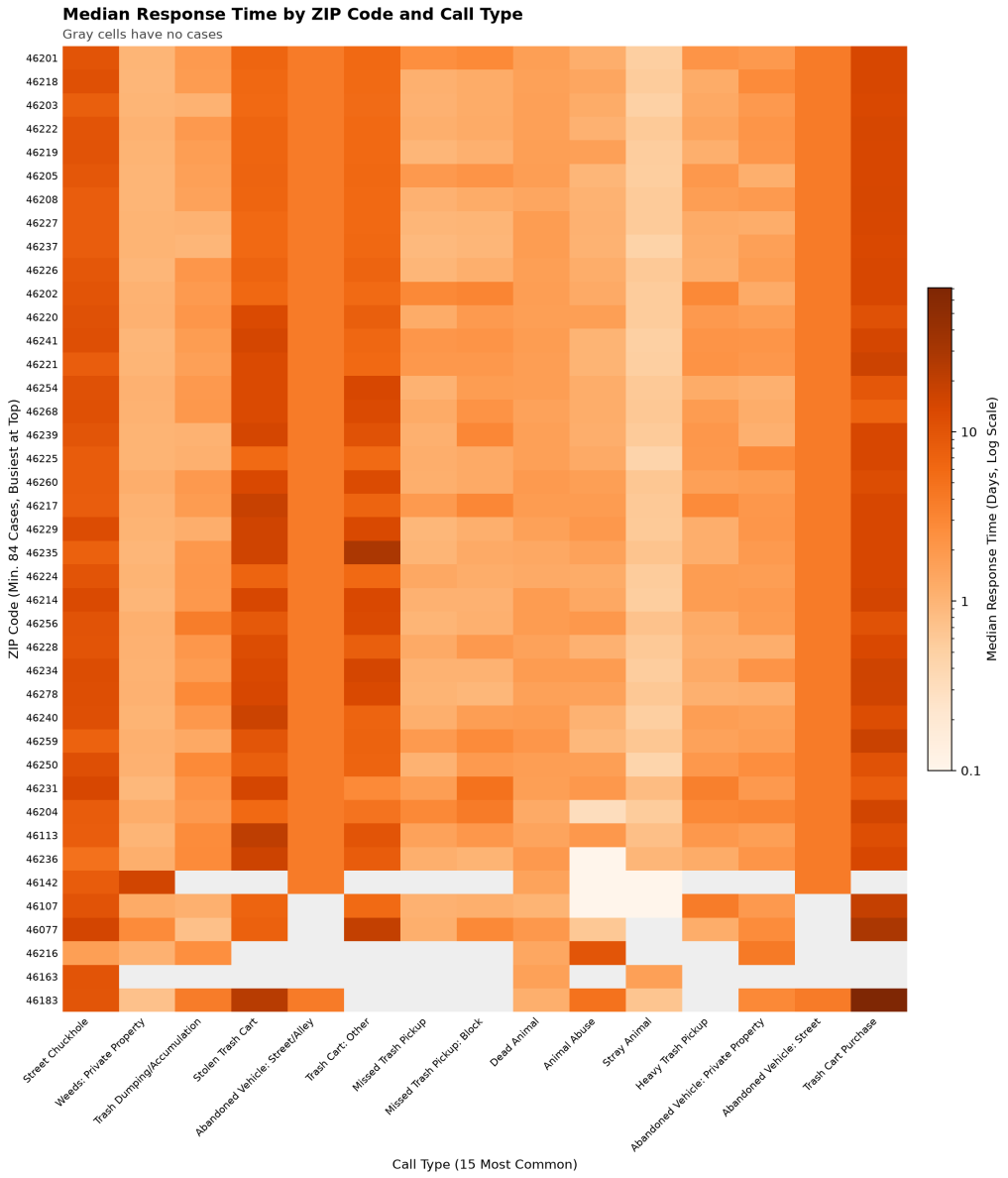


Figure 2. Median response time by ZIP code and request type, for the 15 most common request types (ZIP codes with at least 84 cases). Darker means slower; gray means no cases. Color uses a log scale, in days.

Response time depends more on request type than on location. Each column in the heatmap keeps a similar color from one ZIP code to the next, while colors change much more from one request type to another. Among the 15 most common types, street chuckholes and trash cart requests (stolen carts, other cart issues, and cart purchases) have the longest response times. Any delays an area experiences appear to come from having a disproportionate number of these request types, rather than from its location.

QUESTION 2

What do service request calls look like throughout the year?

> ***Client question.*** *When do different types of requests peak, so the city can plan staffing and resources ahead of time?*

Share of each request type's cases by season

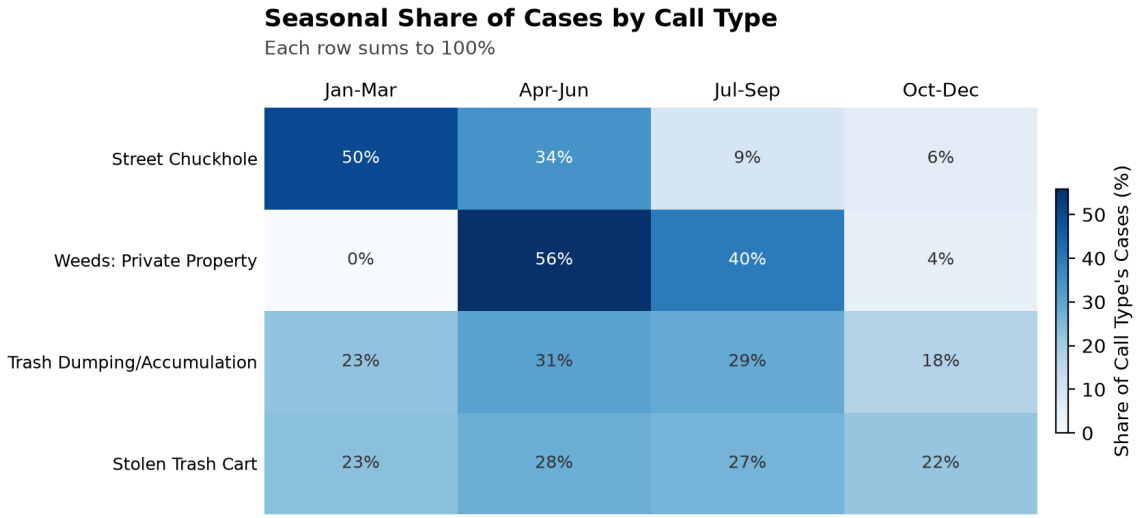


Figure 3. Share of each request type's cases by season (open and closed cases). Each row sums to 100%.

- **Chuckholes and street repairs** are a winter and early spring problem. Half of all cases (50%) arrive from January to March and another 34% from April to June. Only 6% arrive from October to December.
- **Weeds on private property** peak in warm months: 56% of cases come in April to June and 40% in July to September, with almost none in the winter.
- **Trash-related requests** (illegal dumping and stolen carts) are spread much more evenly, between 18% and 31% of cases in any quarter.

Seasonal index

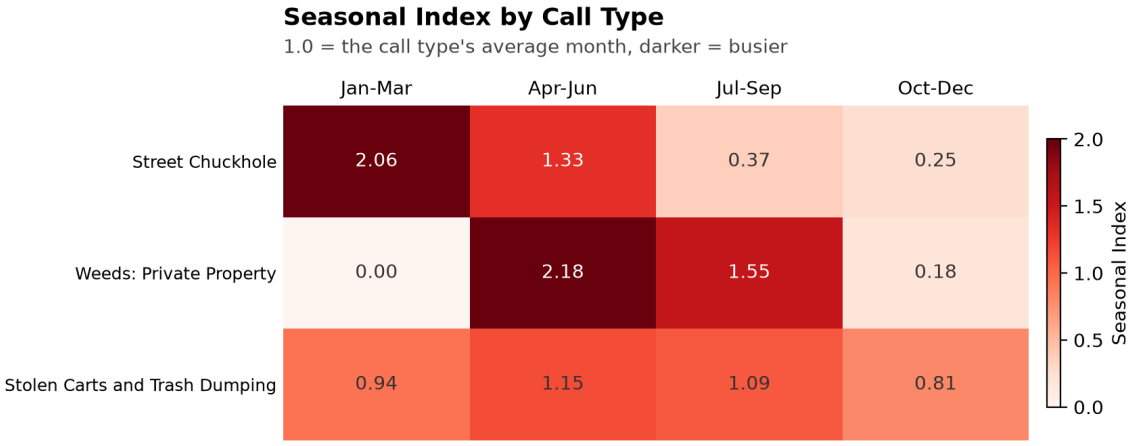


Figure 4. Seasonal index (open and closed cases). A value of 1.0 is an average quarter for that request type, so the four values in each row average 1.0. Values above 1.0 are busier than average; darker means busier.

The index shows the same pattern in terms of workload. Chuckhole calls run at about twice an average quarter (2.06) from January to March, and weeds complaints run at 2.18 from April to June. Trash requests stay between 0.81 and 1.15 all year. October to December is the quietest quarter for all three groups, and trash-related requests hold up best then.

QUESTION 3

What topics and how many calls should the city expect next?

> ***Client question.*** *Where and about what should the city expect requests in upcoming months?*

Method

The forecast for each request type and council district is the average number of cases in the same quarter in prior years. To test it, the first quarter of 2022 was forecast using 2016 to 2021 data and compared with what actually happened. The three most common request types were forecast for their five busiest council districts.

Projected calls by district

Request type	Council district	Projected calls (Q1 2022)
Street Chuckhole	2	916.2
	11	810.3
	1	758.3
	9	751.8
	17	744.5
Trash Dumping/Accumulation	17	239.5
	12	232.3
	11	127.7
	16	118.5
	21	110.7
Stolen Trash Cart	17	259.5
	12	213.3
	11	164.0
	21	122.5
	9	100.5

Table 1. Top five council districts for each of the three most common request types (projected calls = average Q1 cases, 2016 to 2021; open and closed cases).

How accurate was the forecast?

Request type	Projected (Jan + Mar)	Actual (Jan + Mar)	Missed	Count error (Jan + Mar)	Count error (full Q1)
Street Chuckhole	2,437	2,822	571	20.2%	19.6%
Trash Dumping/Accumulation	604	923	319	34.6%	30.0%
Stolen Trash Cart	602	635	46	7.2%	6.2%
Overall	**3,643**	**4,380**	**936**	**21.4%**	**19.7%**

Table 2. Forecast accuracy for Q1 2022 with February excluded (count error = total missed calls divided by total actual calls, for the 15 forecast districts).

What the city should expect

- **Chuckholes, trash dumping, and stolen trash carts** are the main request types to plan for in the first quarter.
- **Council districts 11 and 17** appear in the top five for all three request types, and district 12 appears for both trash types. District 2 has the highest projected chuckhole volume (916 calls), followed by districts 11, 1, 9, and 17.
- **The forecast was off by 19.7% overall** for the full quarter. As a check, February was excluded in case an unusual month distorted the result, and the error stayed similar at 21.4%.
- **Trash dumping is the hardest type to forecast** (30.0% error for the full quarter), while stolen carts are the easiest (6.2%).
- **Projections ran below actual calls** (3,643 projected versus 4,380 actual for January and March), so the projected counts are best treated as a floor. The mix of request types is more dependable than the exact counts, and weather events can add to demand.

The forecast should be used as a planning baseline, not as a precise estimate of call volume.

NEXT STEPS FOR THE CITY

Recommendations

#	Recommendation	When	Where
1	Schedule street maintenance ahead of the winter chuckhole peak, when half of all chuckhole cases arrive.	December, before the January to March peak	Council districts 2, 11, 1, 9, and 17
2	Add capacity for weeds and private property complaints, and offer programs that help homeowners keep up with grass, weeds, and brush.	Before April to June	Citywide
3	Plan trash response, including illegal dumping and stolen carts, around the districts that repeatedly lead the forecast.	Year-round, with a mild rise from April to September	Council districts 17, 12, and 11
4	Target the longest waits. Broken and new sidewalk and uneven street surface requests have median waits of roughly 3 to 4 years for the slowest types. Review the process and funding behind these categories.	Ongoing	Citywide

READING THESE RESULTS

Limitations and next steps

Limitations

- The data reflects issues that residents chose to report. Reporting rates can differ by area, so request counts are not a complete picture of need.
- Response time runs from case creation to case closure. For long-running categories, the closure date may reflect project completion or administrative closure rather than a first response.
- The forecast is a historical average tested on a single quarter. It does not account for weather, population change, or policy changes.
- Request-for-information and switchboard cases were excluded by design.

Suggested next steps

- Test the forecast on additional quarters and compare it with a more flexible model.
- Add community context, such as IndyVitals neighborhood health indicators, to see whether the areas with the longest waits differ in community stability.
- Extend the analysis with the newer RequestIndy data and repeat it at the neighborhood level.

APPENDIX A

Data dictionary

Source. Mayor's Action Center Legacy Service Cases, published on the OpenIndy Data Portal by the City of Indianapolis and Marion County (IndyGIS). The dataset covers June 2015 to August 2022 and comes from the city's legacy case management system. Cases that were open before August 2022 were migrated to the newer *Mayor's Action Center - Request Indy Service Requests* dataset, and closed cases remained in the legacy dataset.

Legacy dataset: hub.arcgis.com/datasets/d2fda038f9c1448a985fb55647593499_3
Newer dataset: data.indy.gov/datasets/IndyGIS::mayors-action-center-request-indy-service-requests

Field	Column in CSV	Description	Used in this report for
(not listed)	`OBJECTID`	Row identifier added by the city's GIS system on export.	Not used
Case Number	`CASENUMBER`	Unique identifier for each service case.	Counting cases
Source ID	`SOURCE_ID__C`	Identifier of the source record for the case.	Not used
Keyword	`KEYWORD__C`	Broad topic of the request.	Request type
Subcategory	`SUBCATEGORY__C`	More specific description of the issue within the keyword.	Request type
Incident Address	`INCIDENT_ADDRESS__C`	Street address where the issue was reported.	Not used
Township	`TOWNSHIP__C`	Marion County township where the issue was reported.	Not used
City	`CITY__C`	City or town where the issue was reported.	Not used
ZIP Code	`ZIP__C`	ZIP code where the issue was reported.	Question 1 (area)
Council District	`COUNCIL_DISTRICT__C`	City-County Council district where the issue was reported.	Question 3 (area)
Record Creation Date	`CREATEDDATE`	Date and time the case was created.	Response time, seasonality
Last Modified Date	`LASTMODIFIEDDATE`	Date and time the case record was last updated.	Not used
Closed Date	`CLOSEDDATE`	Date and time the case was closed.	Response time
Status	`STATUS`	Whether the case is open or closed.	Open and closed cases
Origin	`ORIGIN`	Channel through which the request was received.	Not used

Table A1. Fields in the Mayor's Action Center legacy service cases dataset.

Notes. The Field column gives the portal's display name and the next column gives the header as it appears in the downloaded CSV. The portal lists 14 fields, all stored as text including the date fields; the CSV export adds OBJECTID as the first column. Dates were converted to date-times during cleaning; downloaded timestamps look like `2022/01/31 13:09:51+00` and are in UTC. Response time is Closed Date minus Record Creation Date. Rows missing a field an analysis needs are excluded from that analysis; see Background and approach.